

Geometry
Unit 2
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

For questions 1-3, determine if each of the following represents a definition, postulate, or theorem. Then, draw a picture using the given information.

1. If M is the midpoint of a line segment AB , then $AM = \frac{1}{2}AB$ and $MB = \frac{1}{2}AB$.

Definition

Diagrams may vary.

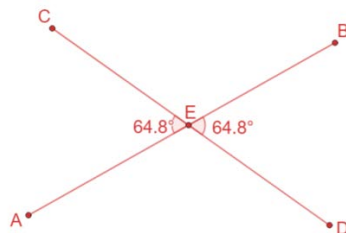


$$AM = \frac{1}{2}AB \text{ and } MB = \frac{1}{2}AB$$

2. Vertical angles are congruent.

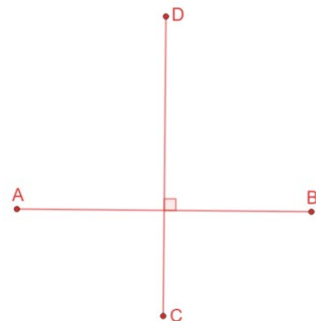
Theorem

Diagrams may vary.



3. Perpendicular lines are lines that intersect to form a right angle.

Definition



Complete the table.

	Property	Equality	Congruence
4.	Transitive	If $m\angle ABC = m\angle DEF$ and $m\angle DEF = 45^\circ$, then $m\angle ABC = 45^\circ$	If $\angle GHI \cong \angle FED$ and $\angle FED \cong \angle ABC$, then $\angle GHI \cong \angle ABC$
5.	Reflexive	$m\angle FGH = m\angle HGF$	$\angle FGH \cong \angle HGF$
6.	Symmetric	If $m\angle XYZ = 19^\circ$, then $19^\circ = m\angle XYZ$	If $\angle XYZ = \angle FGH$, then $\angle FGH = \angle XYZ$

7. Write the conclusion for the statement and name the property.

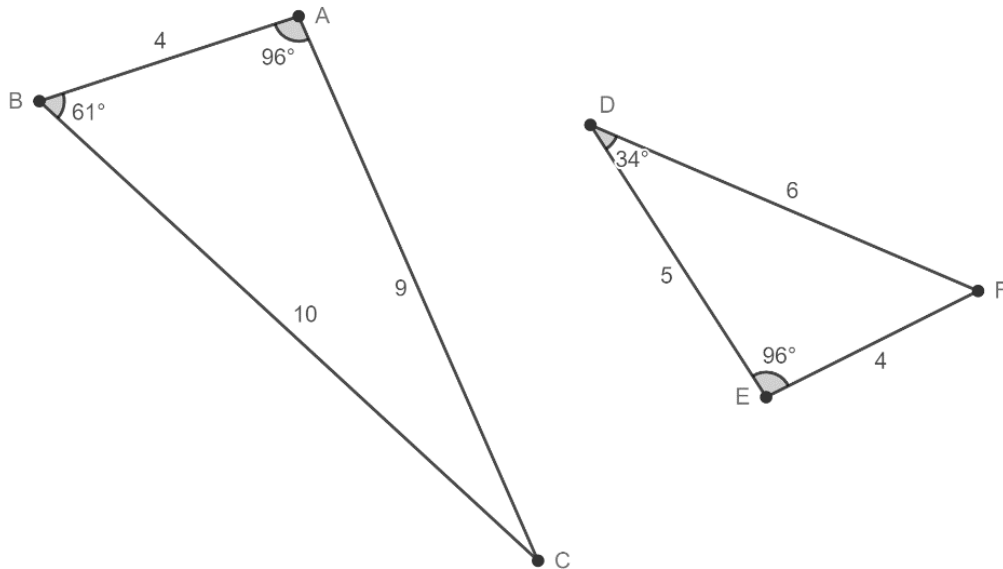
If $m\angle QRS + m\angle LMN = 223^\circ$ and $223^\circ = m\angle GHI$, then...

$m\angle QRS + m\angle LMN = m\angle GHI$; Transitive Property

Complete the table with the reason that makes the given statement true.

	Statement	Reason
8.	$m\angle MNP = m\angle DFG$	Given
9.	$\angle MNP \cong \angle DFG$	Definition of Congruence

10. Triangles ABC and DEF are shown.



Fill in the blanks using the information shown on the two triangles.

$EF = 4$. Therefore, $\overline{EF} \cong \overline{AB}$.

$m\angle CAB = 96^\circ$. Therefore, $\angle CAB \cong \angle DEF$.

Are triangles ABC and DEF congruent? Explain. The triangles are not congruent because corresponding sides are not congruent. (Explanation may vary).

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For questions 1-5, determine the type of reasoning for each statement.

1. For a golf cart to be considered 'street legal' it must have blinkers. Jill's golf cart has been deemed 'street legal'. Therefore, her golf cart has blinkers.

Deductive reasoning

2. When an object is thrown up in the air, it comes back down. Mark throws a baseball up in the air. The baseball will come back down.

Inductive reasoning

3. When Ethan stays up late, he is tired the next day. Ethan stayed up last night until 11:30 pm, therefore, Ethan will be tired the next day.

Deductive reasoning

4. The Olympic games are held every four years. The last games were held in 2020. The next games will be held in 2024.

Deductive reasoning

5. If a student passes an entrance examination, they will be admitted into their local community college. Barry passed the entrance examination. Barry will be admitted into the community college.

Deductive reasoning

For each statement in 6-10, write a counterexample that disproves the statement.

6. If a student attends the junior/senior prom, the student is either a junior or a senior. **Answers may vary. A sophomore attended the prom with a junior.**

7. If someone owns a cell phone, they are able to access the internet from it.

Answers may vary. To protect themselves, some law enforcement officials use cell phones that do not connect to the internet.

8. When two numbers are added, the result will always be a larger number.

Answers may vary. $-3 + -7 = -10$

9. The product of two odd numbers will be even.

Answers may vary. $7 \cdot 11 = 77$

10. If a number is divisible by 2, it is also divisible by 4.

Answers may vary. $\frac{18}{2} = 9$, $\frac{18}{4} = 4.5$, 18 is divisible by 2 and not by 4.

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Lesson 3 Practice

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You and a partner are playing the popular game UNO. You realize there was an error made in one of the steps. Determine which series of steps the error was made in and circle it.

	Statement	Reason
1.	Red 9	Given
2.	Red Skip	Matches color
3.	Yellow Skip	Matches word
4.	Yellow 5	Matches color
5.	Blue 5	Matches number
6.	Green 7 Should be either Green 5	Changes color to green

Complete the proof for the card game UNO.

	Statement	Reason
7.	Red 4	Given
8.	Yellow 4	Changes color to yellow
9.	Yellow Draw Two	Matches Color
10.	Yellow 8	Matches Color

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Lesson 4 Practice

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For questions 1-3, use the quote shown.

"If you look at what you have in life, you'll always have more" – Oprah Winfrey

1. Write the converse of this statement.

"If you always have more, then you'll look at what you have in life."

2. Write the inverse of this statement.

"If you do not look at what you have in life, then you'll never have more."

3. Write the contrapositive of this statement.

"If you will not always have more, then you won't look at what you have in life".

For questions 4-6, use the quote shown.

"The harder I work, the more luck I seem to have" – Thomas Jefferson

4. Write the converse of this statement.

"The more luck I seem to have, the harder I work."

5. Write the inverse of this statement.

"The less I work, the less luck I seem to have."

6. Write the contrapositive of this statement.

"I seem to have less luck, the less I work."

For questions 7-8, use the quote shown.

"If life were predictable it would cease to be life, and be without flavor" – Eleanor Roosevelt

7. What is the hypothesis?

Life was predictable.

8. What is the conclusion?

It would cease to be life and be without flavor.

For questions 9-10, use the quote shown.

"If you're not making mistakes, then you're not doing anything"- John Wooden

9. Write the inverse.

If you are making mistakes, then you are doing something/anything.

10. Write the contrapositive of the quote.

If you are doing anything, then you are making mistakes.

Geometry
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Lesson 5 Practice

Name Answer Key
Date _____
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Complete the table for each given conditional statement.

If a shape is a triangle, then it has exactly three sides.			
	Relationship to Conditional	Statement	True or False
1.	Converse	If a shape has exactly three sides, then it is a triangle.	True
2.	Inverse	If a shape is not a triangle, then it does not have three sides.	True
3.	Contrapositive	If a shape does not have three sides, then it is not a triangle.	True
Write the conditional using "if and only if".			
4.	A shape is a triangle if and only if it has exactly three sides.		

Two angles are supplementary if their sum is 180° .			
	Relationship to Conditional	Statement	True or False
5.	Converse	If the sum of two angles is 180° , then they are supplementary.	True
6.	Inverse	If two angles are not supplementary, then their sum is not 180° .	True
7.	Contrapositive	If the sum of two angles is not 180° , then they are not supplementary.	True
Write the conditional using "if and only if".			
8.	Two angles are supplementary if and only if their sum is 180° .		

9. Write a conditional statement that is biconditional.

Answer will vary.

If a number is rational, then it can be written as a ratio of two integers.

Converse: If a number can be written as a ratio of two integers, it is rational.

Biconditional: A number is rational if and only if it can be written as a ratio of two integers.

10. Write a conditional statement that is **not** biconditional.

Answers will vary.

If you have a temperature over 115° , then you feel fine.

Converse: If you feel fine, then you will have a temperature of 115° . $\rightarrow$ False.

Biconditional: You will have a temperature of 115° if and only if you feel fine. $\rightarrow$ False

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Lesson 6 Practice

Name **Answer Key** _____
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Determine if each statement is true. Rewrite each false statement using the word "some".

	Statement	True or False	Re-write
1.	All odd numbers are composite.	False	Some odd numbers are composite.
2.	All points on a line are collinear.	True	
3.	All even numbers are divisible by 2.	True	

Write each statement as an if-then statement.

	Statement	If-Then
4.	All odd numbers are composite.	If a number is odd, then the number is composite.
5.	All points on a line are collinear.	If points are on the same line, then they are collinear.
6.	All even numbers are divisible by 2.	If a number is even, then the number is divisible by 2.

Write the negation of each statement.

	Statement	Negation
7.	$\triangle TAH$ is an isosceles triangle.	$\triangle TAH$ is not an isosceles triangle.
8.	$\angle BCD$ and $\angle TYG$ are complementary angles.	$\angle BCD$ and $\angle TYG$ are not complementary angles.
9.	$\angle ICH$ and $\angle JNV$ are adjacent angles.	$\angle ICH$ and $\angle JNV$ are not adjacent angles.
10.	$m\angle ABC + m\angle ZYX = 97^\circ$	$m\angle ABC + m\angle ZYX \neq 97^\circ$

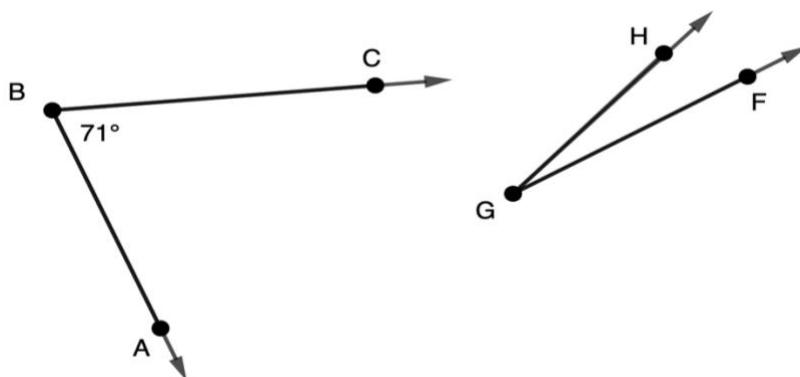
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Lesson 7 Practice

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Complete the two-column proof.

Given: $\angle ABC$ and $\angle HGF$ are complementary; $m\angle ABC = 71^\circ$

Prove: $m\angle HGF = 19^\circ$



	Statement	Reason
1.	$\angle ABC$ and $\angle HGF$ are complementary	Given
2.	$m\angle ABC + m\angle HGF = 90^\circ$	Definition of complementary angles
3.	$m\angle ABC = 71^\circ$	Given
4.	$71^\circ + m\angle HGF = 90^\circ$	Substitution Property
5.	$m\angle HGF = 90^\circ - 71^\circ$ $m\angle HGF = 19^\circ$	Subtraction Property of Equality

Complete the paragraph proof to answer questions 6-10.

Given: $\angle 1$ and $\angle 2$ are supplementary and $\angle 2$ and $\angle 3$ are complementary.

Prove: $\angle 1 \cong \angle 3$

It is given that 6. $\angle 1$ and $\angle 2$ are supplementary and $\angle 2$ and $\angle 3$ are

complementary. By the definition of complementary angles, 7. $m\angle 1 + m\angle 2 = 90^\circ$

and $m\angle 2 + m\angle 3 = 90^\circ$. By the 8. transitive property, $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$.

Then, $m\angle 1 =$ 9. $m\angle 3$ by the subtraction property. Therefore, $\angle 1 \cong \angle 3$ by the

10. definition of congruence

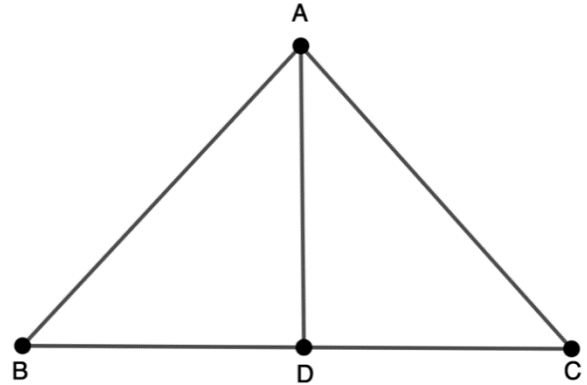
Geometry
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Lesson 8 Practice

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Complete the two-column proof.

Given: $\angle ADB$ is a right angle, $\angle ADB$ and $\angle ADC$ are supplementary.

Prove: $\angle ADB \cong \angle ADC$

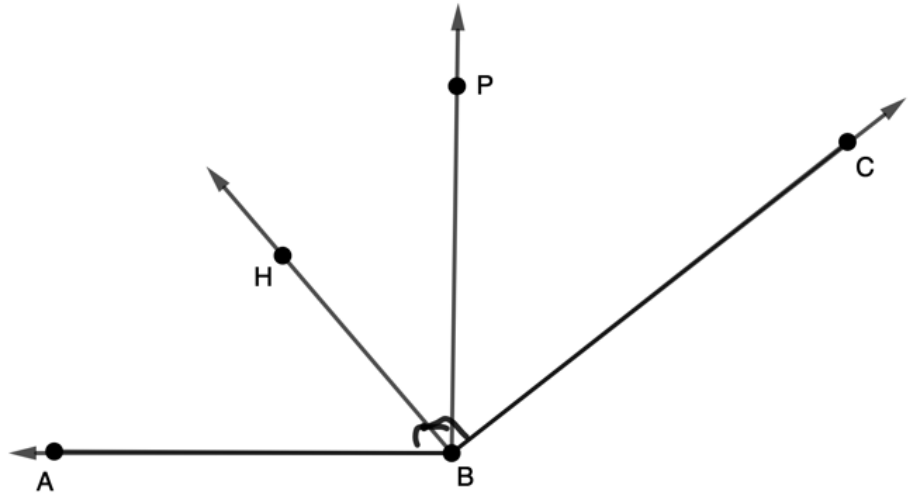


	Statement	Reason
1.	$\angle ADB$ is a right angle	Given
2.	$m\angle ADB = 90^\circ$	Definition of right angle
3.	$\angle ADB$ and $\angle ADC$ are supplementary	Given
4.	$m\angle ADC + m\angle ADB = 180^\circ$	Definition of supplementary angles
5.	$m\angle ADC + 90^\circ = 180^\circ$	Substitution Property
6.	$m\angle ADC = 90^\circ$	Subtraction Property
7.	$m\angle ADC = m\angle ADB$	Transitive Property of Equality
8.	$\angle ADC \cong \angle ADB$	Definition of Congruence

Complete the paragraph proof to answer questions 5-10.

Given: $\angle PBA$ and $\angle HBC$ are right angles

Prove: $\angle ABH \cong \angle PBC$



It is given that $\angle PBA$ and $\angle HBC$ are right angles. Using the addition postulate,

6. $m\angle ABH + m\angle HBP = m\angle PBA$ and 7. $m\angle HBP + m\angle PBC = m\angle HBC$.

Using the 8. transitive property of equality, we know

$m\angle ABH + m\angle HBP = m\angle HBP + m\angle PBC$. Next, we apply the subtraction property of equality

to determine 9. $m\angle ABH = m\angle PBC$. By the definition of 10. congruence,

$\angle ABH \cong \angle PBC$.